Detailed Solutions to Practice Problems on Non-linear Regression

Advanced Statistical Methods II

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Abstract

This document gives self-contained, step-by-step solutions to all fourteen problems in the attached Practice Problems on Non-linear Regression sheet. Each original question is reproduced in full before the solution. The required theory is stated before each derivation, with explicit explanations of the transformations, identifiability arguments, numerical calculations, optimization steps, and inferential conclusions. The model definitions and the Gauss–Newton, Newton, and Fisher-scoring formulas are taken from the supplied PS4 sheet; no outside course material is introduced.

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Preliminary theory: derivation of the nonlinear least-squares algorithms

The PS4 sheet explicitly asks the reader to derive Gauss–Newton, Newton, and Fisher scoring before using them. We do so here, using the notation of the sheet. These derivations also explain why the algorithms have different update formulas.

Common notation

Let

$$f(\theta) = \begin{pmatrix} f(x_1, \theta) \\ \vdots \\ f(x_n, \theta) \end{pmatrix}, \qquad r(\theta) = y - f(\theta), \qquad S(\theta) = r(\theta)^T r(\theta).$$

The Jacobian $J(\theta)$ is the $n \times p$ matrix with entries

$$J_{ij}(\theta) = \frac{\partial f(x_i, \theta)}{\partial \theta_j}.$$

A small parameter increment is denoted by δ, so the next iterate is $\theta^{(k+1)} = \theta^{(k)} + \delta$.

Tool A: Gauss–Newton update

Step 1: Linearize the mean function. A first-order Taylor expansion around the current iterate gives

$$f(\theta^{(k)} + \delta) \approx f(\theta^{(k)}) + J(\theta^{(k)})\delta.$$

Therefore the residual vector is approximately

$$r(\theta^{(k)} + \delta) \approx r(\theta^{(k)}) - J(\theta^{(k)})\delta.$$

Step 2: Replace the nonlinear problem by least squares in δ. The local approximation to the objective is

$$S(\theta^{(k)} + \delta) \approx \|r - J\delta\|^2.$$

This is an ordinary least-squares problem with response r and design matrix J.

Step 3: Solve its normal equations. Differentiating with respect to δ gives

$$-2J^T(r - J\delta) = 0 \quad \Longleftrightarrow \quad J^T J\delta = J^T r.$$

When $J^T J$ is invertible,

$$\delta = (J^T J)^{-1} J^T r,$$

so

$$\boxed{\theta^{(k+1)} = \theta^{(k)} + (J^T J)^{-1} J^T r.}$$

For one parameter, J is a column with entries $J_i = f_i'(\theta)$, giving

$$\boxed{\theta^{(k+1)} = \theta^{(k)} + \frac{\sum_i J_i r_i}{\sum_i J_i^2}.}$$

The essential approximation is to linearize the mean function and then solve the resulting least-squares problem.

Tool B: Newton update

Step 1: Differentiate the exact objective. Since $r_i = y_i - f_i(\theta)$,

$$\nabla S(\theta) = -2J^T r.$$

A second differentiation gives

$$\nabla^2 S(\theta) = 2J^T J - 2\sum_{i=1}^{n} r_i(\theta)\nabla^2 f(x_i, \theta).$$

Step 2: Minimize the second-order Taylor approximation. Around $\theta^{(k)}$, write

$$S(\theta^{(k)} + \delta) \approx S(\theta^{(k)}) + \nabla S(\theta^{(k)})^T \delta + \frac{1}{2}\delta^T \nabla^2 S(\theta^{(k)})\delta.$$

The stationary point of this quadratic approximation solves

$$\nabla^2 S(\theta^{(k)})\delta = -\nabla S(\theta^{(k)}).$$

Substitute the gradient and Hessian and divide by 2:

$$\left(J^T J - \sum_i r_i \nabla^2 f_i\right)\delta = J^T r.$$

If the matrix is positive definite (so the quadratic stationary point is a minimum),

$$\boxed{\theta^{(k+1)} = \theta^{(k)} + \left(J^T J - \sum_i r_i \nabla^2 f_i\right)^{-1} J^T r.}$$

For one parameter this becomes

$$\boxed{\theta^{(k+1)} = \theta^{(k)} + \frac{\sum_i J_i r_i}{\sum_i J_i^2 - \sum_i r_i f_i''}.}$$

Unlike Gauss–Newton, Newton retains the residual-weighted curvature of the nonlinear mean function. That can improve the local approximation, but it can also make a finite step unstable or undefined.

Tool C: Fisher scoring and its relation to Gauss–Newton/IRLS

For a log-likelihood $\ell(\theta)$, Newton's method for maximization uses the observed information $-\nabla^2 \ell(\theta)$:

$$\theta^{(k+1)} = \theta^{(k)} + \{-\nabla^2 \ell(\theta^{(k)})\}^{-1} \nabla \ell(\theta^{(k)}).$$

Fisher scoring replaces the observed information by its expectation,

$$I(\theta) = \mathbb{E}\{-\nabla^2 \ell(\theta)\},$$

so the update is

$$\boxed{\theta^{(k+1)} = \theta^{(k)} + I(\theta^{(k)})^{-1} \nabla \ell(\theta^{(k)}).}$$

For normal nonlinear regression with independent $N(0, \sigma^2)$ errors,

$$\ell(\theta) = -\frac{1}{2\sigma^2} S(\theta) + \text{constant}.$$

The score is $J^T r/\sigma^2$. In the observed information, the residual-curvature term has expectation zero because $\mathbb{E}(r_i \mid x_i) = 0$. Thus

$$I(\theta) = \frac{1}{\sigma^2} J^T J,$$

and Fisher scoring gives exactly the Gauss–Newton increment $(J^T J)^{-1} J^T r$.

For a generalized linear model, let $\eta_i = x_i^T \beta$, $\mu_i = g^{-1}(\eta_i)$, and $\mathrm{Var}(Y_i) = \phi V(\mu_i)$. A first-order change in the mean gives

$$d\mu_i = g'(\mu_i)^{-1} d\eta_i,$$

so the Fisher-scoring/IRLS calculation can be written as weighted least squares of the working response

$$z_i = \eta_i + (y_i - \mu_i) g'(\mu_i)$$

with weights

$$W_{ii} = \frac{1}{V(\mu_i) g'(\mu_i)^2},$$

all evaluated at the current iterate. The update is

$$\boxed{\beta^{(k+1)} = (X^T W^{(k)} X)^{-1} X^T W^{(k)} z^{(k)}.}$$

The factor $1/\phi$ cancels from the coefficient update. This is the IRLS formula printed on the PS4 sheet.

What the three methods approximate

Gauss–Newton minimizes a least-squares objective after a first-order linearization of the mean. Newton minimizes a second-order Taylor approximation to the actual objective. Fisher scoring replaces observed curvature in a likelihood problem by expected curvature. Their steps can coincide in special models, but in general they are different; Newton's potentially rapid local convergence does not by itself provide a global decrease guarantee.

1 Problem 1: Linearity in the parameters and possible transformations

Problem Statement

For each mean function decide: (i) is the model linear in the parameters (use the Jacobian criteria: the model is linear if and only if every $\partial f/\partial\theta_j$ is free of θ)? (ii) Can a transformation make it linear, and under which error structure? (iii) Is there any other trouble?

(a) $f(x,\theta) = \theta_1 + \theta_2 x + \theta_3 x^2$.

(b) $f(x,\theta) = \theta_1 x^{\theta_2}$, $x > 0$.

(c) $f(x,\theta) = \theta_1 + \theta_2 e^{\theta_3 x}$.

(d) $f(x,\theta) = (\theta_1 + \theta_2 x)^2$.

(e) $f(x,\theta) = \dfrac{\theta_1 x}{\theta_2 + x}$.

Required Theory

A regression model is linear in its parameters when the mean function can be expressed as a linear combination of known functions of the predictor, with the unknown parameters entering only as coefficients. The Jacobian criterion supplied in the problem sheet is that every partial derivative $\partial f/\partial\theta_j$ must be free of the unknown parameter vector. A nonlinear transformation of the response may make the displayed mean relation algebraically linear, but it also changes the error distribution. Therefore one must distinguish:

1. a transformation of the mean equation, which may be algebraically valid; and
2. a transformation of the statistical model, which is valid only when the error structure is compatible with that transformation.

For example, a logarithm converts a multiplicative error into an additive error on the log scale, while taking reciprocals of an additive-error response does not preserve an additive, constant-variance error model.

Solution

(a) Polynomial in x.

The mean is

$$f(x,\theta) = \theta_1 \cdot 1 + \theta_2 \cdot x + \theta_3 \cdot x^2.$$

The unknown parameters appear only as coefficients of known functions. The derivatives are

$$\frac{\partial f}{\partial\theta_1} = 1, \qquad \frac{\partial f}{\partial\theta_2} = x, \qquad \frac{\partial f}{\partial\theta_3} = x^2,$$

none of which depends on θ. Thus the model is linear in the parameters. No response transformation is needed. The main practical issue is design identifiability: the columns $1, x, x^2$ must be linearly independent on the observed design. In particular, at least three distinct predictor values are needed to identify all three coefficients, and near-collinearity among these columns can make estimation unstable.

(b) Power model.

The derivatives are

$$\frac{\partial f}{\partial \theta_1} = x^{\theta_2}, \qquad \frac{\partial f}{\partial \theta_2} = \theta_1 x^{\theta_2} \log x.$$

They depend on unknown parameters, so the model is nonlinear in (θ_1, θ_2).

If $\theta_1 > 0$ and the mean is positive, take logarithms:

$$\log f(x, \theta) = \log \theta_1 + \theta_2 \log x.$$

Set $\eta_1 = \log \theta_1$ and $z = \log x$. Then

$$\log f = \eta_1 + \theta_2 z,$$

which is linear in (η_1, θ_2). This transformation is statistically natural when the original noise is multiplicative, for example

$$Y = \theta_1 x^{\theta_2} e^{\varepsilon}, \quad \text{so that} \quad \log Y = \eta_1 + \theta_2 \log x + \varepsilon.$$

If instead the original model is $Y = f(x, \theta) + \varepsilon$ with additive errors of constant variance, taking logs changes the error to $\log(f + \varepsilon) - \log f$, which is neither additive nor generally constant-variance. Ordinary least squares after logging is then a different fitting criterion. The log transformation also requires positive observations and a positive mean.

(c) Exponential component plus an unknown offset.

Here

$$\frac{\partial f}{\partial \theta_1} = 1, \qquad \frac{\partial f}{\partial \theta_2} = e^{\theta_3 x}, \qquad \frac{\partial f}{\partial \theta_3} = \theta_2 x e^{\theta_3 x}.$$

The last two derivatives depend on unknown parameters, so the model is nonlinear. If θ_1 were known and $Y - \theta_1$ were positive, then

$$\log\{f(x, \theta) - \theta_1\} = \log \theta_2 + \theta_3 x,$$

which is linear in $(\log \theta_2, \theta_3)$. This transformation requires a multiplicative error structure for the component $Y - \theta_1$, not the usual additive error around the full mean. When θ_1 is unknown, the unknown offset must first be estimated, so a direct logarithmic linearization is unavailable. Degenerate cases such as $\theta_2 = 0$ or $\theta_3 = 0$ also make parameters difficult or impossible to separate.

(d) Square of an affine function.

The derivatives are

$$\frac{\partial f}{\partial \theta_1} = 2(\theta_1 + \theta_2 x), \qquad \frac{\partial f}{\partial \theta_2} = 2x(\theta_1 + \theta_2 x),$$

which depend on the unknown parameters. Therefore the model is nonlinear.

Algebraically, if the mean is positive and its square-root branch is known, one can write

$$\sqrt{f(x, \theta)} = \theta_1 + \theta_2 x$$

or its negative counterpart. The sign matters: (θ_1, θ_2) and $(-\theta_1, -\theta_2)$ give exactly the same squared mean, so the original parameters are not globally identifiable without a sign convention or other restriction. Moreover, applying a square root to observations with additive errors does not produce an ordinary additive constant-variance error model. A linear fit to $\sqrt{Y}$ is statistically equivalent only under an appropriate error assumption on the transformed scale.

(e) Michaelis–Menten-type rational mean.

The derivatives are

$$\frac{\partial f}{\partial \theta_1} = \frac{x}{\theta_2 + x}, \qquad \frac{\partial f}{\partial \theta_2} = -\frac{\theta_1 x}{(\theta_2 + x)^2},$$

so the model is nonlinear in its parameters. For $x \neq 0$ and a nonzero mean, taking reciprocals gives

$$\frac{1}{f(x,\theta)} = \frac{\theta_2 + x}{\theta_1 x} = \frac{1}{\theta_1} + \frac{\theta_2}{\theta_1}\frac{1}{x}.$$

This is a straight-line relation between $1/f$ and $1/x$. The reciprocal relation is algebraically exact for the mean function, and can be fitted as an exact linear relationship when the transformed quantities are treated as error-free or when the model is explicitly specified with additive error on the reciprocal scale. If the original observations have additive errors $Y = f + \varepsilon$, then $1/Y$ has a nonlinear, heteroskedastic error and this transformation does not preserve the original least-squares problem. This distinction is the reason a reciprocal plotting method need not agree with nonlinear least squares in a noisy sample. The domain must also avoid $\theta_2 + x = 0$.

Key Takeaway

A mean function can be algebraically transformed into a linear relation without the transformed regression being statistically equivalent to the original model. Always check the error structure, parameter identifiability, and domain as well as the algebra.

2 Problem 2: Identifiability: what can the data determine?

Problem Statement

For each model, which parameters (or functions of them) can be estimated? Suggest a fix.

(a) $f(x,\theta) = \theta_1 e^{\theta_2 + \theta_3 x}$.

(b) $f(x,\theta) = \theta_1(1 - e^{-\theta_2 x})$, with $\theta_1, \theta_2 > 0$, when

  (i) every observation is taken at the same $x_0 > 0$;

  (ii) observations are taken at two distinct values $0 < x_a < x_b$. Is θ now identifiable? Does the NLSE always exist?

(c) $f(x,\theta) = \theta_1 e^{-\theta_2 x} + \theta_3 e^{-\theta_4 x}$.

(d) $f(x,\theta) = \beta \cos(\theta x)$, observed at $x = 1, 2, \ldots, n$.

Required Theory

A parameter is identifiable if different parameter values cannot produce the same mean function on the design under consideration. A parameterization may fail to be identifiable even when a lower-dimensional function of its parameters is identifiable. Identifiability is a property of the model and design, not merely of whether an optimizer returns a numerical answer.

For nonlinear least squares, the objective is

$$S(\theta) = \sum_{i=1}^{n} \{y_i - f(x_i,\theta)\}^2.$$

An NLSE exists if this objective attains its infimum at an allowed finite parameter value. If the allowed mean-function set is open and the best approximation lies only on its boundary, the infimum may not be attained.

Solution

(a) Product confounding.

Rewrite the mean as

$$f(x,\theta) = \underbrace{(\theta_1 e^{\theta_2})}_{A} e^{\theta_3 x}.$$

The data see $A = \theta_1 e^{\theta_2}$ and θ_3, not θ_1 and θ_2 separately. Indeed, for any real c,

$$\theta_1^* = \theta_1 e^c, \qquad \theta_2^* = \theta_2 - c$$

gives the same product $\theta_1^* e^{\theta_2^*} = \theta_1 e^{\theta_2}$. Thus there is an entire one-dimensional family of parameter values producing the same mean. A natural fix is to reparameterize directly in terms of $A = \theta_1 e^{\theta_2}$ and θ_3, or to fix one of θ_1, θ_2 using external information.

(b1) All observations at one positive design value.

If every $x_i = x_0 > 0$, then every fitted mean equals

$$\mu_0 = \theta_1(1 - e^{-\theta_2 x_0}).$$

The data can estimate this single mean value, but they cannot separate θ_1 from θ_2. For any chosen $\theta_2 > 0$, one can set

$$\theta_1 = \frac{\mu_0}{1 - e^{-\theta_2 x_0}}$$

to obtain exactly the same fitted mean (when $\mu_0 > 0$). Repeating observations at the same x_0 improves estimation of μ_0, but it does not resolve the confounding. The fix is to observe at multiple distinct positive predictor values.

(b2) Two distinct positive design values.

Let the two fitted group means be

$$m_a = \theta_1(1 - e^{-\theta_2 x_a}), \qquad m_b = \theta_1(1 - e^{-\theta_2 x_b}).$$

Taking their ratio eliminates θ_1:

$$R(\theta_2) = \frac{m_b}{m_a} = \frac{1 - e^{-\theta_2 x_b}}{1 - e^{-\theta_2 x_a}}.$$

As $\theta_2 \downarrow 0$, the approximation $1 - e^{-\theta_2 x} \sim \theta_2 x$ gives

$$R(\theta_2) \longrightarrow \frac{x_b}{x_a}.$$

As $\theta_2 \to \infty$, both exponentials tend to zero, so $R(\theta_2) \to 1$. Moreover, R is strictly decreasing. To see this, differentiate its logarithm:

$$\frac{d}{d\theta_2} \log R(\theta_2) = \frac{x_b}{e^{\theta_2 x_b} - 1} - \frac{x_a}{e^{\theta_2 x_a} - 1} < 0,$$

because for fixed positive θ_2 the function $x \mapsto x/(e^{\theta_2 x} - 1)$ is decreasing. Hence R maps $(0, \infty)$ one-to-one onto $(1, x_b/x_a)$. If the two mean values satisfy

$$m_a > 0, \qquad 1 < \frac{m_b}{m_a} < \frac{x_b}{x_a},$$

there is a unique $\theta_2 > 0$ solving the ratio equation, and then

$$\theta_1 = \frac{m_a}{1 - e^{-\theta_2 x_a}}$$

is uniquely determined. Thus two distinct positive design points remove the structural non-identifiability for a mean pair inside the model's range.

Does an NLSE always exist? No. The attainable pair (m_a, m_b) must satisfy strict inequalities. For example, suppose the two observed group means are both equal to 1. The model predicts $m_b > m_a$ for every finite $\theta_2 > 0$, but can approach equal means as $\theta_2 \to \infty$ with θ_1 near 1. The least-squares infimum can therefore be approached at a boundary that no finite parameter reaches. This example shows that identifiability does not guarantee existence of an NLSE. A practical fix is to ensure design and data are compatible with the parameter constraints, or to use a closed parameter space with an explicitly handled boundary model.

(c) Sum of two exponentials.

The model is

$$f(x, \theta) = \theta_1 e^{-\theta_2 x} + \theta_3 e^{-\theta_4 x}.$$

Interchanging the two parameter pairs leaves the mean unchanged:

$$(\theta_1, \theta_2, \theta_3, \theta_4) \quad \text{and} \quad (\theta_3, \theta_4, \theta_1, \theta_2)$$

produce exactly the same function. Thus the labels of the two components are not identifiable. If $\theta_2 = \theta_4$, then the mean reduces to $(\theta_1 + \theta_3)e^{-\theta_2 x}$, so only the sum of the amplitudes is identifiable. A standard fix is to impose an ordering restriction such as $\theta_2 < \theta_4$, use positive amplitudes/rates when scientifically appropriate, and collect data at sufficiently many well-spread predictor values. The ordering removes label switching but cannot separate components whose rates are actually equal.

(d) Cosine with integer design points.

The means are $\beta \cos(\theta x)$ for integer $x = 1, \ldots, n$. Since

$$\cos\{(\theta + 2\pi k)x\} = \cos(\theta x), \qquad \cos\{(-\theta)x\} = \cos(\theta x)$$

for every integer k, the data cannot distinguish θ from $\theta + 2\pi k$ or $-\theta$. Thus global identifiability of θ fails on the unrestricted real line. If $\beta = 0$, all fitted means are zero and θ is completely unidentifiable. A fix is to restrict θ to a scientifically meaningful interval that removes periodic and sign aliases, require a nonzero amplitude, and use enough distinct design points so that the remaining mean pattern identifies the parameters locally.

Key Takeaway

More observations at the same design point cannot resolve structural confounding. Identifiability depends on the shape of the model and the design, while existence of a minimizer is a separate question about whether the least-squares infimum is attained.

3 Problem 3: A linear model in disguise: a cubic parameter

Problem Statement

Data: $x = (1, 2, 3)$, $y = (2.1, 3.9, 6.2)$. Model: $Y_i = \theta^3 x_i + \varepsilon_i$, $\theta \in \mathbb{R}$.

(a) Find $\widehat{\theta}$ in closed form. Does it always exist? Is it unique?

(b) Show that the Gauss–Newton update is

$$\theta^{(k+1)} = \theta^{(k)} + \frac{\widehat{\beta} - \{\theta^{(k)}\}^3}{3\{\theta^{(k)}\}^2},$$

where $\widehat{\beta}$ is the least-squares slope of a line through the origin. It is Newton's method for solving which equation?

(c) Starting from $\theta^{(0)} = 1$, compute two iterations. What happens from $\theta^{(0)} = 0$? From $\theta^{(0)} = -1$?

(d) Show that $\sum_i f_i''(\widehat{\theta}) r_i(\widehat{\theta}) = 0$ here. What does this imply for the speed of Gauss–Newton (see Problem 11)?

Required Theory

The nonlinear least-squares estimator minimizes

$$S(\theta) = \sum_i \{y_i - f(x_i, \theta)\}^2.$$

A useful idea in the supplied problem sheet is a "linear model in disguise": if $f(x, \theta) = g(\theta)x$ and g is one-to-one, first estimate $\beta = g(\theta)$ by ordinary through-origin least squares and then invert g.

For one parameter, define $J_i = f_i'(\theta)$ and $r_i = y_i - f_i(\theta)$. The Gauss–Newton update is

$$\theta^{(k+1)} = \theta^{(k)} + \frac{\sum_i J_i r_i}{\sum_i J_i^2},$$

whereas the Newton update for minimizing S is

$$\theta^{(k+1)} = \theta^{(k)} + \frac{\sum_i J_i r_i}{\sum_i J_i^2 - \sum_i r_i f_i''}.$$

Solution

(a) Closed-form estimate and existence.

Set $\beta = \theta^3$. The model becomes the through-origin linear regression $Y_i = \beta x_i + \varepsilon_i$. Its least-squares slope is

$$\widehat{\beta} = \frac{\sum_i x_i y_i}{\sum_i x_i^2} = \frac{1(2.1) + 2(3.9) + 3(6.2)}{1^2 + 2^2 + 3^2} = \frac{2.1 + 7.8 + 18.6}{14} = \frac{28.5}{14} = \frac{57}{28}.$$

Since the cube function $g(\theta) = \theta^3$ maps $\mathbb{R}$ one-to-one onto $\mathbb{R}$, the unique minimizer is

$$\boxed{\widehat{\theta} = \left(\frac{57}{28}\right)^{1/3} \approx 1.26738.}$$

It always exists for any finite data because $\widehat{\beta}$ is finite and every real number has a unique real cube root. It is unique because the map $\theta \mapsto \theta^3$ is one-to-one.

(b) Derive the Gauss–Newton step and identify Newton's equation.

Here

$$f_i(\theta) = \theta^3 x_i, \qquad J_i(\theta) = 3\theta^2 x_i, \qquad r_i(\theta) = y_i - \theta^3 x_i.$$

Substitute these into Gauss–Newton:

$$\begin{aligned}
\theta^{(k+1)} &= \theta^{(k)} + \frac{\sum_i 3\{\theta^{(k)}\}^2 x_i (y_i - \{\theta^{(k)}\}^3 x_i)}{\sum_i 9\{\theta^{(k)}\}^4 x_i^2} \\
&= \theta^{(k)} + \frac{\sum_i x_i y_i - \{\theta^{(k)}\}^3 \sum_i x_i^2}{3\{\theta^{(k)}\}^2 \sum_i x_i^2} \\
&= \boxed{\theta^{(k)} + \frac{\widehat{\beta} - \{\theta^{(k)}\}^3}{3\{\theta^{(k)}\}^2}}.
\end{aligned}$$

The last expression is exactly Newton's method for finding a root of

$$h(\theta) = \theta^3 - \widehat{\beta} = 0,$$

because $h'(\theta) = 3\theta^2$ and

$$\theta_{\text{new}} = \theta - \frac{\theta^3 - \widehat{\beta}}{3\theta^2}.$$

(c) Numerical iterations and starting values.

The value used in the update is $\widehat{\beta} = 57/28 \approx 2.0357143$.

Starting from $\theta^{(0)} = 1$,

$$\theta^{(1)} = 1 + \frac{2.0357143 - 1}{3} = 1.3452381.$$

The second update is

$$\theta^{(2)} = 1.3452381 + \frac{2.0357143 - (1.3452381)^3}{3(1.3452381)^2} \approx \boxed{1.2717960}.$$

This moves toward the exact value 1.2673764.

If $\theta^{(0)} = 0$, the denominator $3\{\theta^{(0)}\}^2$ is zero. The Gauss–Newton/Newton formula is undefined. This is not a numerical rounding issue: the derivative of θ^3 is zero at zero, so the local linearization contains no first-order information about how to move.

If $\theta^{(0)} = -1$, the first iterate is

$$\theta^{(1)} = -1 + \frac{2.0357143 - (-1)}{3} \approx 0.0119048.$$

This is very close to zero, where the derivative $3\theta^2$ is extremely small. The next Newton step therefore becomes enormous:

$$\theta^{(2)} \approx 0.0119048 + \frac{2.0357143 - (0.0119048)^3}{3(0.0119048)^2} \approx 4.7880 \times 10^3.$$

Thus a perfectly finite initial value can produce a dramatic overshoot. The example warns that Newton-type updates depend strongly on the starting value and on the local derivative.

(d) The residual-curvature term vanishes at the estimate.

The second derivative is

$$f_i''(\theta) = 6\theta x_i.$$

At the estimate,

$$\sum_i f_i''(\widehat{\theta}) r_i(\widehat{\theta}) = 6\widehat{\theta} \sum_i x_i \{y_i - \widehat{\theta}^3 x_i\}.$$

But $\widehat{\theta}^3 = \widehat{\beta}$, and the through-origin OLS normal equation gives

$$\sum_i x_i (y_i - \widehat{\beta} x_i) = 0.$$

Therefore

$$\boxed{\sum_i f_i''(\widehat{\theta}) r_i(\widehat{\theta}) = 0.}$$

By the convergence formula in Problem 11, the derivative of the Gauss–Newton iteration map at the solution is zero. Hence the local first-order error term vanishes: Gauss–Newton is faster than ordinary linear convergence here (under the usual smoothness conditions, it has at least quadratic local behavior).

Key Takeaway

The parameterization θ^3 hides a simple linear regression, but the iterative algorithm can still behave badly near a point where its derivative vanishes. The exact estimator exists uniquely even though a particular Newton starting value can be problematic.

4 Problem 4: A closed-form estimator that may fail to exist

Problem Statement

Let $Y_i = e^\theta + \varepsilon_i$, $i = 1, \ldots, n$, with ε_i i.i.d. $N(0, \sigma^2)$.

(a) Find the NLSE $\widehat{\theta}$. When does it exist? Compute the probability that it does not exist when $\theta = 0$, $\sigma = 2$, $n = 4$. (You may use $\Phi(-1) = 0.1587$.)

(b) Is $\widehat{\theta}$ unbiased? Use a second-order expansion to approximate its bias and variance.

(c) Evaluate the approximations for $e^\theta = 5$, $\sigma = 2$, $n = 4$. A simulation with 4×10^6 samples gives bias -0.0212 and standard deviation 0.212. Comment.

(d) Compare your variance with the general formula $\sigma^2 (J^T J)^{-1}$.

(e) Is $e^{\widehat{\theta}}$ an unbiased estimator of e^θ?
What does this say about bias and parametrization?

Required Theory

For a nonlinear mean $f(x_i, \theta)$, the NLSE minimizes $S(\theta) = \sum_i \{y_i - f(x_i, \theta)\}^2$. Here the fitted mean $\mu = e^\theta$ is constrained to be strictly positive. For a normal sample with known variance, the least-squares estimator of an unrestricted common mean is the sample mean $\overline{Y}$.

For a smooth transformation g, a second-order Taylor expansion around a random variable's mean gives the delta-method approximations

$$g(\overline{Y}) \approx g(\mu) + g'(\mu)(\overline{Y} - \mu) + \frac{1}{2} g''(\mu)(\overline{Y} - \mu)^2.$$

Since $\mathbb{E}(\overline{Y} - \mu) = 0$ and $\mathrm{Var}(\overline{Y}) = \sigma^2/n$, this yields approximate bias and variance. The general local variance formula supplied in the problem sheet is $\sigma^2 (J^T J)^{-1}$.

Solution

(a) Closed form, existence, and probability of non-existence.

Because all fitted means equal $e^\theta = \mu$,

$$S(\theta) = \sum_{i=1}^{n} (Y_i - e^\theta)^2.$$

For an unrestricted real common mean μ, the minimizer is $\overline{Y}$. But here $\mu = e^\theta$ must be positive. Therefore:

- If $\overline{Y} > 0$, the minimizing fitted mean is $e^{\widehat{\theta}} = \overline{Y}$, so $\widehat{\theta} = \log \overline{Y}$.

- If $\overline{Y} \leq 0$, no finite θ gives the unconstrained minimizing mean. The objective approaches its infimum as $e^\theta \downarrow 0$ (that is, $\theta \to -\infty$), but the value zero is not attained by e^θ for any finite θ. Hence a finite NLSE does not exist.

Thus

$$\boxed{\widehat{\theta} = \log \overline{Y} \quad \text{if and only if } \overline{Y} > 0.}$$

Since $\overline{Y} \sim N(e^\theta, \sigma^2/n)$,

$$\mathbb{P}(\text{no finite NLSE}) = \mathbb{P}(\overline{Y} \leq 0) = \Phi\left(-\frac{e^\theta \sqrt{n}}{\sigma}\right).$$

For $\theta = 0$, $e^\theta = 1$, $\sigma = 2$, $n = 4$:

$$\mathbb{P}(\text{no finite NLSE}) = \Phi\left(-\frac{1 \cdot 2}{2}\right) = \Phi(-1) = \boxed{0.1587.}$$

(b) Bias and variance by a second-order expansion.

Let $\mu = e^\theta$ and $\Delta = \overline{Y} - \mu$. On the event where the estimator exists,

$$\widehat{\theta} = \log(\mu + \Delta).$$

Expand $g(y) = \log y$ around μ:

$$\log(\mu + \Delta) \approx \log \mu + \frac{\Delta}{\mu} - \frac{\Delta^2}{2\mu^2}.$$

Taking expectations and using $\mathbb{E}\Delta = 0$, $\mathbb{E}\Delta^2 = \sigma^2/n$ gives

$$\text{Bias}(\widehat{\theta}) \approx -\frac{1}{2\mu^2}\frac{\sigma^2}{n} = \boxed{-\frac{\sigma^2}{2ne^{2\theta}}}.$$

For the first-order variance, retain the linear term:

$$\widehat{\theta} - \theta \approx \frac{\Delta}{\mu},$$

so

$$\text{Var}(\widehat{\theta}) \approx \frac{1}{\mu^2}\,\text{Var}(\overline{Y}) = \boxed{\frac{\sigma^2}{ne^{2\theta}}}.$$

The logarithm is concave, which is consistent with the negative approximate bias. Strictly, the estimator is undefined on $\{\overline{Y} \leq 0\}$; these delta-method calculations are local approximations in the regime where that event has small probability.

(c) Numerical evaluation and comparison with simulation.

Here $e^\theta = \mu = 5$, $\sigma = 2$, $n = 4$. Thus

$$\text{Bias}(\widehat{\theta}) \approx -\frac{2^2}{2 \cdot 4 \cdot 5^2} = -\frac{1}{50} = \boxed{-0.0200},$$

and

$$\text{Var}(\widehat{\theta}) \approx \frac{2^2}{4 \cdot 5^2} = 0.0400, \qquad \text{SD}(\widehat{\theta}) \approx \boxed{0.2000}.$$

The reported simulation gives bias -0.0212 and standard deviation 0.212. These are close to the approximations. The remaining discrepancy is expected: the Taylor approximation ignores

higher-order terms and the small probability of a non-existent estimate. At these parameter values,

$$\mathbb{P}(\overline{Y} \le 0) = \Phi(-5\sqrt{4}/2) = \Phi(-5),$$

which is extremely small, so the existence issue has very little effect on the simulation in this case.

(d) Compare with $\sigma^2(J^TJ)^{-1}$.

Each fitted mean is $f_i(\theta) = e^\theta$, so

$$J_i = \frac{\partial f_i}{\partial \theta} = e^\theta = \mu.$$

Therefore

$$J^TJ = \sum_{i=1}^{n} J_i^2 = n\mu^2 = ne^{2\theta},$$

and the supplied general variance formula gives

$$\sigma^2(J^TJ)^{-1} = \frac{\sigma^2}{ne^{2\theta}},$$

which exactly matches the first-order delta-method variance derived in part (b).

(e) Bias depends on the parameterization.

Whenever the NLSE exists,

$$e^{\widehat{\theta}} = e^{\log \overline{Y}} = \overline{Y}.$$

The sample mean $\overline{Y}$ is unbiased for $\mu = e^\theta$, because $\mathbb{E}\overline{Y} = \mu$. This illustrates that an estimator can be biased for θ while the corresponding estimator on the original mean scale is unbiased: bias is not invariant under nonlinear transformations of a parameter.

There is one important qualification in this problem. Under the positive parameterization $\mu = e^\theta$, the finite NLSE does not exist when $\overline{Y} \le 0$. Thus $e^{\widehat{\theta}}$ is not defined on those samples unless an extra convention is introduced. The algebraic identity above and the unbiasedness of $\overline{Y}$ apply to the natural estimator of μ; they do not remove the non-existence event for the log-parameter estimator.

Key Takeaway

The mean parameter $\mu = e^\theta$ has the unbiased estimator $\overline{Y}$, but the transformed estimator $\log \overline{Y}$ is biased and is not even finite on every sample. Parameterization affects both bias and existence.

5 Problem 5: Saturated designs and closed-form nonlinear least squares

Problem Statement

(a) Suppose the design has exactly p distinct values $x_1, \ldots, x_p$, with n_k observations and mean $\overline{y}_k$ at x_k. Show that

$$S(\theta) = \sum_{k=1}^{p}\sum_{i=1}^{n_k}(y_{ki} - \overline{y}_k)^2 + \sum_{k=1}^{p} n_k\{\overline{y}_k - f(x_k, \theta)\}^2,$$

and deduce: if $f(x_k, \theta^*) = \overline{y}_k$ for all k, then θ^* is an NLSE.

(b) $f(x,\theta) = \theta_1 e^{-\theta_2 x}$, with observations at $x = 0$ and $x = c$ only. Find the NLSE in closed form. When does it exist? Evaluate it for $\overline{y}_0 = 100$, $\overline{y}_c = 40$, $c = 2$ hours.

(c) Michaelis–Menten model $f(x; V_m, K) = V_m x/(K+x)$, using only the Puromycin (treated) runs at $x = 0.02$ (rates $76, 47$) and $x = 1.10$ (rates $207, 200$). Find $\widehat{V}_m$ and $\widehat{K}$ in closed form.

(d) The Lineweaver–Burk method (OLS of $1/y$ on $1/x$) generally gives biased estimates of V_m and K. Why is it exact here?

Required Theory

For observations grouped at the same design value, write $y_{ki} = \overline{y}_k + (y_{ki} - \overline{y}_k)$. The within-group deviations sum to zero, so the cross term vanishes when a squared residual is expanded. This yields the standard decomposition into within-group variation and lack of fit of the fitted mean to the group means.

A design with p distinct predictor values and p parameters is called saturated when the model can match all p group means. If it does, the second (lack-of-fit) term is zero, and no other parameter value can reduce the first (within-group) term.

Solution

(a) Prove the saturated-design decomposition.

At the kth design point, write $f_k = f(x_k, \theta)$. Then

$$y_{ki} - f_k = (y_{ki} - \overline{y}_k) + (\overline{y}_k - f_k).$$

Squaring and summing over the replicates at x_k gives

$$\sum_{i=1}^{n_k}(y_{ki} - f_k)^2 = \sum_i (y_{ki} - \overline{y}_k)^2 + 2(\overline{y}_k - f_k)\sum_i (y_{ki} - \overline{y}_k) + n_k(\overline{y}_k - f_k)^2.$$

By the definition of the sample mean,

$$\sum_i (y_{ki} - \overline{y}_k) = 0.$$

The cross term therefore vanishes. Summing over k proves

$$\boxed{S(\theta) = \sum_{k=1}^{p}\sum_{i=1}^{n_k}(y_{ki} - \overline{y}_k)^2 + \sum_{k=1}^{p} n_k\{\overline{y}_k - f(x_k,\theta)\}^2.}$$

The first term is independent of θ and nonnegative. If θ^* matches every group mean, the second term is zero, its smallest possible value. Hence θ^* is an NLSE.

(b) Exponential decay at two design points.

The two fitted group means satisfy

$$f(0,\theta) = \theta_1, \qquad f(c,\theta) = \theta_1 e^{-\theta_2 c}.$$

To interpolate the observed group means, set

$$\theta_1 = \overline{y}_0, \qquad \overline{y}_c = \overline{y}_0 e^{-\theta_2 c}.$$

Provided the ratio is positive and the parameters are allowed to take the corresponding values,

$$\boxed{\widehat{\theta}_1 = \overline{y}_0, \qquad \widehat{\theta}_2 = -\frac{1}{c}\log\left(\frac{\overline{y}_c}{\overline{y}_0}\right).}$$

Under the usual exponential-decay restrictions $\theta_1 > 0$ and $\theta_2 > 0$, a finite exact fit exists precisely when

$$\boxed{0 < \overline{y}_c < \overline{y}_0.}$$

If the parameters are instead unrestricted real numbers, the exact-interpolation condition is that $\overline{y}_0 \neq 0$ and $\overline{y}_c/\overline{y}_0 > 0$ (with the special all-zero pair fitted by $\theta_1 = 0$ if that boundary is allowed). The problem statement does not separately write the parameter space, so the positive-decay interpretation is the natural one for the numerical application.

For the supplied values,

$$\widehat{\theta}_1 = 100, \qquad \widehat{\theta}_2 = -\frac{1}{2}\log(40/100) = \frac{1}{2}\log(2.5) \approx \boxed{0.458145 \text{ hour}^{-1}}.$$

The model matches both group means exactly, so the saturated-design decomposition proves that these values form the NLSE.

(c) Michaelis–Menten estimates.

First compute the two group means:

$$\overline{y}_1 = \frac{76 + 47}{2} = 61.5, \qquad \overline{y}_2 = \frac{207 + 200}{2} = 203.5,$$

with $x_1 = 0.02$ and $x_2 = 1.10$. Because there are two parameters and two distinct design values, an exact fit to these two group means gives the NLSE whenever the resulting parameters belong to the intended parameter space.

The interpolation equations are

$$61.5 = \frac{V_m(0.02)}{K + 0.02}, \qquad 203.5 = \frac{V_m(1.10)}{K + 1.10}.$$

Rearrange each equation to express V_m:

$$V_m = 61.5\frac{K + 0.02}{0.02} = 203.5\frac{K + 1.10}{1.10}.$$

Equating these expressions and solving for K gives

$$K = \frac{x_1 x_2(\overline{y}_2 - \overline{y}_1)}{\overline{y}_1 x_2 - \overline{y}_2 x_1}.$$

Substitution yields

$$K = \frac{(0.02)(1.10)(203.5 - 61.5)}{(61.5)(1.10) - (203.5)(0.02)} = \frac{3.124}{63.58} \approx \boxed{0.049135}.$$

Then

$$V_m = 61.5\frac{0.04913495 + 0.02}{0.02} \approx \boxed{212.590}.$$

Both estimates are positive, and substituting them into the two mean equations reproduces 61.5 and 203.5. The within-group sums of squares are fixed, and the lack-of-fit term is zero; hence these are the NLSEs for the two selected design groups.

(d) Why Lineweaver–Burk is exact here.

Taking reciprocals of the Michaelis–Menten mean gives

$$\frac{1}{f(x)} = \frac{K + x}{V_m x} = \frac{1}{V_m} + \frac{K}{V_m}\frac{1}{x}.$$

Thus, if we first replace the two replicate groups by their two original-scale means $\overline{y}_1$ and $\overline{y}_2$, and then regress $1/\overline{y}_k$ on $1/x_k$, the two-point straight line interpolates both transformed means exactly. Its intercept and slope are $1/\widehat{V}_m$ and $\widehat{K}/\widehat{V}_m$, and back-transformation gives the same closed-form parameters as part (c). This is the saturated two-point calculation.

There is an important distinction if "OLS of $1/y$ on $1/x$" is interpreted literally as transforming all four individual rates before fitting. That regression fits the two group means of the reciprocals, $\overline{(1/y)}_k$, not the reciprocals of the original group means, $1/\overline{y}_k$. In general,

$$\overline{(1/y)}_k \neq \frac{1}{\overline{y}_k},$$

so this individual-observation Lineweaver–Burk fit is not the same as the original-scale NLSE, even though the transformed regression has only two distinct predictor values. For the four supplied rates, the individual-reciprocal fit gives approximately $\widehat{V}_m = 213.327$ and $\widehat{K} = 0.05346$, rather than the original-scale NLSE values 212.590 and 0.049135. Therefore the exactness claim is valid for the two-group-mean transformed calculation; it is not literally valid for the standard procedure applied separately to all four replicate observations. The distinction comes from the nonlinear reciprocal transformation, not from a failure of the algebraic straight-line identity.

Key Takeaway

When the number of distinct design points matches the number of freely adjustable parameters and the model can interpolate the group means, nonlinear least squares has a closed form. A transformed straight-line method is exact in this example because both procedures interpolate the same two group means, not because reciprocal transformations are generally harmless.

6 Problem 6: The three-point method for asymptotic regression

Problem Statement

Asymptotic regression: $y(x) = \alpha + (y_0 - \alpha)e^{-kx}$, $k > 0$. There are three equally spaced design points $x, x + h, x + 2h$, with means $\overline{y}_1, \overline{y}_2, \overline{y}_3$.

(a) Show that $y(x + h) - \alpha = \rho\{y(x) - \alpha\}$ with $\rho = e^{-kh}$. Deduce the closed forms

$$\widehat{\rho} = \frac{\overline{y}_3 - \overline{y}_2}{\overline{y}_2 - \overline{y}_1}, \qquad \widehat{k} = -\frac{\log \widehat{\rho}}{h}, \qquad \widehat{\alpha} = \frac{\overline{y}_1\overline{y}_3 - \overline{y}_2^2}{\overline{y}_1 + \overline{y}_3 - 2\overline{y}_2}.$$

Why is this the NLSE?

(b) A cup of coffee reads $90, 70, 56°C$ at $t = 0, 5, 10$ minutes. Estimate the room temperature and the cooling rate, and predict the reading at $t = 15$.

(c) When does the solution exist? Try the biochemical oxygen demand (BOD) data at days $1, 2, 3$ $(8.3, 10.3, 19.0)$ and at days $3, 4, 5$ $(19.0, 16.0, 15.6)$.

(d) Why is the method still useful when there are many data points?

Required Theory

For the asymptotic model, moving the predictor forward by h multiplies the difference from the asymptote by e^{-kh}. With $\rho = e^{-kh}$ and $k > 0$, the ratio must satisfy $0 < \rho < 1$. The three equally spaced model means therefore form a geometric progression after subtracting α:

$$y_2 - \alpha = \rho(y_1 - \alpha), \qquad y_3 - \alpha = \rho(y_2 - \alpha).$$

There are three unknown quantities (α, y_0, k) and three distinct design-point means, so if the equations have an admissible solution the model interpolates all three means and the saturated-design decomposition establishes NLSE optimality.

Solution

(a) Derive the three closed forms.

Start with

$$y(x) = \alpha + (y_0 - \alpha)e^{-kx}.$$

At $x + h$,

$$\begin{aligned} y(x+h) - \alpha &= (y_0 - \alpha)e^{-k(x+h)} \\ &= e^{-kh}(y_0 - \alpha)e^{-kx} \\ &= e^{-kh}\{y(x) - \alpha\}. \end{aligned}$$

Thus $\rho = e^{-kh}$ and

$$y_2 - \alpha = \rho(y_1 - \alpha), \qquad y_3 - \alpha = \rho(y_2 - \alpha).$$

Subtract consecutive equations:

$$y_2 - y_1 = (\rho - 1)(y_1 - \alpha),$$

$$y_3 - y_2 = (\rho - 1)(y_2 - \alpha).$$

Since $y_2 - \alpha = \rho(y_1 - \alpha)$, division gives

$$\boxed{\rho = \frac{y_3 - y_2}{y_2 - y_1}.}$$

Replacing the model values by the observed group means gives $\widehat{\rho}$. Now solve $\rho = e^{-kh}$:

$$\boxed{\widehat{k} = -\frac{1}{h}\log\widehat{\rho}.}$$

To obtain α, use $y_2 - \alpha = \rho(y_1 - \alpha)$ and substitute the ratio expression. Rearranging gives

$$\boxed{\alpha = \frac{y_1 y_3 - y_2^2}{y_1 + y_3 - 2y_2}.}$$

Replacing the model values by the observed means yields the stated $\widehat{\alpha}$.

Why is this the NLSE? There are three parameters and three distinct design points. If the formulas give an admissible finite parameter triple, then the model fits all three group means exactly. By Problem 5(a), the second (lack-of-fit) term is then zero, which is the smallest possible value. Therefore the interpolating triple is an NLSE.

(b) Coffee temperature.

Here $h = 5$, and

$$\overline{y}_1 = 90, \qquad \overline{y}_2 = 70, \qquad \overline{y}_3 = 56.$$

The geometric ratio is

$$\widehat{\rho} = \frac{56 - 70}{70 - 90} = \frac{-14}{-20} = 0.7.$$

Therefore

$$\widehat{k} = -\frac{1}{5}\log(0.7) \approx \boxed{0.071335 \text{ min}^{-1}}.$$

The estimated room temperature is

$$\widehat{\alpha} = \frac{90(56) - 70^2}{90 + 56 - 2(70)} = \frac{5040 - 4900}{6} = \boxed{23.333°\text{C}}.$$

At 15 minutes, the temperature difference from the room temperature is multiplied by another factor of 0.7:

$$\widehat{y}(15) = \widehat{\alpha} + (56 - \widehat{\alpha})(0.7) = 23.333 + 32.667(0.7) = \boxed{46.2°\text{C}}.$$

(c) Existence conditions and the BOD triples.

For a finite positive cooling/approach rate $k > 0$, we need

$$0 < \widehat{\rho} < 1.$$

We also need $\overline{y}_2 - \overline{y}_1 \neq 0$ so the ratio is defined, and the denominator

$$\overline{y}_1 + \overline{y}_3 - 2\overline{y}_2$$

for $\widehat{\alpha}$ must be nonzero. Once $0 < \widehat{\rho} < 1$, the fitted amplitude is determined by the first point. If the ratio is negative, zero, or at least one, no finite $k > 0$ interpolates the three group means.

For days 1, 2, 3 the means are $(8.3, 10.3, 19.0)$. The ratio is

$$\widehat{\rho} = \frac{19.0 - 10.3}{10.3 - 8.3} = \frac{8.7}{2.0} = 4.35 > 1.$$

This would give $\widehat{k} = -\log(4.35) < 0$, which violates $k > 0$. Thus the three-point asymptotic model cannot interpolate this triple with an admissible positive rate.

For days 3, 4, 5 the means are $(19.0, 16.0, 15.6)$. Now

$$\widehat{\rho} = \frac{15.6 - 16.0}{16.0 - 19.0} = \frac{-0.4}{-3.0} = 0.133333\ldots \in (0, 1).$$

Hence

$$\widehat{k} = -\log(0.133333\ldots) \approx \boxed{2.0149 \text{ day}^{-1}},$$

and

$$\widehat{\alpha} = \frac{19.0(15.6) - 16.0^2}{19.0 + 15.6 - 32.0} = \frac{40.4}{2.6} \approx \boxed{15.5385}.$$

This triple has an admissible exact interpolating fit.

(d) Why use it with many observations?

With more than three design points, the three-point formulas no longer use all the information and are not generally the full NLSE. They are still useful for generating initial values (α, y_0, k) for a nonlinear least-squares routine. A good starting value can improve convergence and reduce the risk that an iterative method starts in a poor region. The final fit should then use all observations.

Key Takeaway

The three-point method is exact when its ratio is between zero and one and the denominators are nonzero. With many data points, its main role is to provide principled starting values for full nonlinear least squares.

7 Problem 7: Profiling out a linear parameter

Problem Statement

Consider $Y_i = \theta_1 g_i(\theta_2) + \varepsilon_i$ with

$$g_i(\theta_2) = 1 - e^{-\theta_2 x_i}$$

(the BOD model). Write $g(\theta_2) = (g_1, \ldots, g_n)^T$ and $y = (y_1, \ldots, y_n)^T$.

(a) For fixed θ_2, find $\widehat{\theta}_1(\theta_2)$ minimizing $S(\theta_1, \theta_2)$.

(b) Show that the profile sum of squares is

$$\widetilde{S}(\theta_2) = S(\widehat{\theta}_1(\theta_2), \theta_2) = \|y\|^2 - \frac{(y^T g)^2}{\|g\|^2} = \|y\|^2 \sin^2 \alpha(\theta_2),$$

where $\alpha(\theta_2)$ is the angle between y and $g(\theta_2)$. Interpret $\widehat{\theta}_2$ geometrically.

(c) Show that

$$\frac{d\widetilde{S}}{d\theta_2} = \frac{\partial S}{\partial \theta_2}\Big|_{\theta_1 = \widehat{\theta}_1(\theta_2)},$$

and conclude that minimizing $\widetilde{S}$ gives the NLSE.

(d) Why is this useful in practice?

Required Theory

Profiling means minimizing an objective with respect to one parameter while holding the remaining parameter(s) fixed, and then minimizing the resulting lower-dimensional objective. Here θ_1 is linear once θ_2 is fixed, so its conditional least-squares estimate is available by a one-variable OLS calculation. The envelope identity follows from the chain rule and the fact that the derivative of S with respect to the profiled parameter is zero at its conditional minimizer.

Solution

(a) Minimize over θ_1 for fixed θ_2.

For fixed θ_2, abbreviate $g = g(\theta_2)$. The model vector is $\theta_1 g$, so

$$S(\theta_1, \theta_2) = \|y - \theta_1 g\|^2 = (y - \theta_1 g)^T (y - \theta_1 g).$$

Expand:

$$S = y^T y - 2\theta_1 y^T g + \theta_1^2 g^T g.$$

Differentiate with respect to θ_1:

$$\frac{\partial S}{\partial \theta_1} = -2y^T g + 2\theta_1 g^T g.$$

Setting the derivative equal to zero gives

$$\boxed{\widehat{\theta}_1(\theta_2) = \frac{y^T g(\theta_2)}{g(\theta_2)^T g(\theta_2)}},$$

provided $g(\theta_2) \neq 0$. The second derivative is $2g^T g > 0$, so this is the unique conditional minimizer.

(b) Derive and interpret the profile objective.

Substitute the conditional minimizer into the expanded sum of squares:

$$\begin{aligned}\widetilde{S}(\theta_2) &= y^T y - 2\frac{y^T g}{g^T g} y^T g + \left(\frac{y^T g}{g^T g}\right)^2 g^T g \\ &= y^T y - \frac{(y^T g)^2}{g^T g}.\end{aligned}$$

Therefore

$$\boxed{\widetilde{S}(\theta_2) = \|y\|^2 - \frac{(y^T g)^2}{\|g\|^2}.}$$

Let $\alpha(\theta_2)$ denote the angle between the vectors y and $g(\theta_2)$. By the inner-product identity

$$y^T g = \|y\| \, \|g\| \cos\alpha,$$

we obtain

$$\widetilde{S}(\theta_2) = \|y\|^2\{1 - \cos^2\alpha(\theta_2)\} = \boxed{\|y\|^2 \sin^2\alpha(\theta_2).}$$

Geometrically, for each fixed θ_2, the best $\theta_1 g(\theta_2)$ is the orthogonal projection of y onto the one-dimensional direction spanned by $g(\theta_2)$. The profile residual sum of squares is the squared length of the component perpendicular to that direction. Thus $\widehat{\theta}_2$ chooses the direction $g(\theta_2)$ that makes the angle with y as small (or as close to zero, up to the sign allowed by θ_1) as possible.

(c) Envelope identity and global minimization.

The profile objective is a composition:

$$\widetilde{S}(\theta_2) = S(\widehat{\theta}_1(\theta_2), \theta_2).$$

Apply the chain rule:

$$\frac{d\widetilde{S}}{d\theta_2} = \frac{\partial S}{\partial \theta_1}\frac{d\widehat{\theta}_1}{d\theta_2} + \frac{\partial S}{\partial \theta_2},$$

where the partial derivatives are evaluated at $(\widehat{\theta}_1(\theta_2), \theta_2)$. By part (a), the first-order condition for the conditional minimum is

$$\left.\frac{\partial S}{\partial \theta_1}\right|_{\theta_1 = \widehat{\theta}_1(\theta_2)} = 0.$$

The first chain-rule term therefore disappears, leaving

$$\boxed{\frac{d\widetilde{S}}{d\theta_2} = \left.\frac{\partial S}{\partial \theta_2}\right|_{\theta_1 = \widehat{\theta}_1(\theta_2)}}.$$

For every fixed θ_2, part (a) has already chosen the best possible θ_1. Minimizing the resulting profile over θ_2 therefore minimizes the original sum of squares over both parameters:

$$\min_{\theta_2} \widetilde{S}(\theta_2) = \min_{\theta_2} \min_{\theta_1} S(\theta_1, \theta_2) = \min_{\theta_1, \theta_2} S(\theta_1, \theta_2).$$

Thus the minimizing pair is the NLSE, subject to the same existence and parameter-space restrictions as the original problem.

(d) Practical usefulness.

Profiling reduces a two-parameter nonlinear optimization problem to a one-dimensional optimization problem. The inner parameter θ_1 is solved exactly at every trial value of θ_2, so an iterative routine need only search over θ_2. This reduces the numerical dimension, avoids unnecessary iteration over a parameter that has a closed form, and often makes the optimization more stable and interpretable.

Key Takeaway

Whenever some parameters enter linearly conditional on the others, solve those parameters analytically first. Profiling removes them from the numerical search without changing the global least-squares problem.

8 Problem 8: Logistic regression with a binary covariate: Challenger O-rings

Problem Statement

The Challenger O-ring data, with launches split at 65°F:

	flights	O-rings at risk	O-rings distressed
cold (< 65°F)	4	24	5
warm (≥ 65°F)	19	114	4

Model: $\operatorname{logit} \pi = \beta_0 + \beta_1 \cdot \text{cold}$, with binomial counts.

(a) Show that the MLE has a closed form: $\widehat{\pi}_g = k_g/m_g$ in each group g, where k_g is the number distressed and m_g is the number at risk. Find $\widehat{\beta}_0$ and $\widehat{\beta}_1$, and interpret $e^{\widehat{\beta}_1}$.

(b) Show that the estimated variance of $\widehat{\beta}_1$ is

$$\frac{1}{k_c} + \frac{1}{m_c - k_c} + \frac{1}{k_w} + \frac{1}{m_w - k_w}.$$

Give a 95% interval for the odds ratio ($z_{0.975} = 1.96$).

(c) In what sense is this a saturated design, as in Problem 5?

(d) With temperature as a continuous covariate, the same model gives $\widehat{\beta}_1 = -0.1156$ per °F.
What is lost by dichotomizing temperature?

Required Theory

For independent binomial groups,

$$K_g \sim \text{Binomial}(m_g, \pi_g),$$

the group log-likelihood is

$$\ell_g(\pi_g) = k_g \log \pi_g + (m_g - k_g) \log(1 - \pi_g) + \text{constant}.$$

Its maximizer is the sample proportion $\widehat{\pi}_g = k_g/m_g$. In a two-group logistic model, the warm group is the reference group and the cold coefficient is the difference between the two group logits. The odds ratio is the exponential of that logit difference.

For a binomial proportion, the large-sample variance of its logit estimate is approximately $1/k + 1/(m-k)$. Since the difference of independent group logits has variance equal to the sum of their variances, the displayed variance formula follows.

Solution

(a) Closed-form MLEs and odds ratio.

Let c denote cold and w denote warm. The two sample proportions are

$$\widehat{\pi}_c = \frac{5}{24} = 0.208333, \qquad \widehat{\pi}_w = \frac{4}{114} = \frac{2}{57} \approx 0.035088.$$

The model is

$$\text{logit}(\pi_w) = \beta_0, \qquad \text{logit}(\pi_c) = \beta_0 + \beta_1.$$

At the MLE, each group-specific fitted probability equals its binomial sample proportion. Therefore

$$\widehat{\beta}_0 = \text{logit}(\widehat{\pi}_w) = \log\left(\frac{4}{110}\right) \approx \boxed{-3.31419},$$

and

$$\widehat{\beta}_1 = \text{logit}(\widehat{\pi}_c) - \text{logit}(\widehat{\pi}_w) = \log\left(\frac{5}{19}\right) - \log\left(\frac{4}{110}\right) \approx \boxed{1.97918}.$$

Exponentiating gives the cold-to-warm odds ratio:

$$e^{\widehat{\beta}_1} = \frac{5/19}{4/110} = \frac{550}{76} = \frac{275}{38} \approx \boxed{7.23684}.$$

The estimated odds of O-ring distress are about 7.24 times as large in the cold group as in the warm group under this two-level model.

(b) Variance and 95% interval.

The estimated variance of the cold-group logit is

$$\widehat{\text{Var}}\{\text{logit}(\widehat{\pi}_c)\} = \frac{1}{5} + \frac{1}{19}.$$

For the warm group it is

$$\widehat{\text{Var}}\{\text{logit}(\widehat{\pi}_w)\} = \frac{1}{4} + \frac{1}{110}.$$

The groups are independent, so the variance of their difference is the sum:

$$\begin{aligned}\widehat{\text{Var}}(\widehat{\beta}_1) &= \frac{1}{5} + \frac{1}{19} + \frac{1}{4} + \frac{1}{110} \\ &\approx \boxed{0.511722}.\end{aligned}$$

Thus $\text{se}(\widehat{\beta}_1) = \sqrt{0.511722} \approx 0.715348$. The 95% Wald interval on the log-odds-ratio scale is

$$1.979185 \pm 1.96(0.715348) = (0.577103, 3.381267).$$

Exponentiate the two endpoints to obtain the interval for the odds ratio:

$$\boxed{\left(e^{0.577103}, e^{3.381267}\right) \approx (1.781, 29.408).}$$

The interval is wide because the number of distressed rings is small, especially in the warm group.

(c) Why this is saturated.

There are two distinct covariate groups and two logistic parameters, β_0 and β_1. The model can choose one probability for the warm group and a separate probability for the cold group. It therefore matches both observed group proportions exactly:

$$\widehat{\pi}_w = 4/114, \qquad \widehat{\pi}_c = 5/24.$$

There is no lack-of-fit degree of freedom at the level of the two group proportions, just as in the saturated-design discussion of Problem 5. The binomial sampling variation within each group remains, but the two group means/proportions are exactly represented.

(d) Information lost by dichotomizing temperature.

Replacing temperature by a cold/warm indicator discards the actual temperature differences within each group. A launch at 34°F and a launch at 64°F are treated as identical, as are launches at 65°F and 80°F. The fitted model estimates a single jump at the chosen cutoff rather than a continuous temperature-response relationship. Consequently, dichotomization loses within-group ordering and information about the shape and rate of change of risk with temperature; it can also make the result sensitive to the selected threshold.

Key Takeaway

The binary-covariate logistic model is saturated at the two group proportions, so the MLE is available in closed form. Dichotomization makes that fit simple but discards the information carried by the original continuous temperatures.

9 Problem 9: Poisson regression: likelihood and scoring

Problem Statement

$Y_i \sim \text{Poisson}(\mu_i)$ independently, with $\log \mu_i = \eta_i = x_i^T \beta$.

(a) Write down the log-likelihood, the score, the observed information $-\nabla^2 \ell$, and the expected information. Show that the last two coincide.

(b) Show that the Newton (equivalently, Fisher scoring) update is a weighted least-squares fit of a working response z_i on x_i, with weights w_i. Find z_i and w_i.

(c) Data: $x = (0, 1, 2)$, $y = (2, 3, 6)$, model $\log \mu = \beta_0 + \beta_1 x$. Starting from $\beta^{(0)} = (\log \overline{y}, 0)$, compute $\beta^{(1)}$ by hand.

(d) Why can there be no spurious local maxima here, unlike in nonlinear least squares?

(e) Show that at the MLE, $\sum_i y_i = \sum_i \widehat{\mu}_i$.

Required Theory

For a Poisson random variable,

$$\Pr(Y_i = y_i) = \frac{e^{-\mu_i} \mu_i^{y_i}}{y_i!}.$$

With $\mu_i = e^{x_i^T \beta}$, the log-likelihood is a concave function of β. Its score is the design matrix transpose times the residual vector $y - \mu$. Newton's method updates a parameter by multiplying the score by the inverse information matrix. In a generalized linear model, this update can be written as weighted least squares of a working response on the design matrix.

Solution

(a) Likelihood, score, and information.

The joint likelihood is

$$L(\beta) = \prod_{i=1}^{n} \frac{e^{-\mu_i} \mu_i^{y_i}}{y_i!}.$$

Taking logarithms and using $\log \mu_i = x_i^T \beta$ gives

$$\ell(\beta) = \sum_i \{y_i x_i^T \beta - e^{x_i^T \beta} - \log(y_i!)\}.$$

Differentiate with respect to β:

$$\boxed{\nabla \ell(\beta) = \sum_i x_i (y_i - \mu_i) = X^T (y - \mu).}$$

Differentiating again,

$$\nabla^2 \ell(\beta) = -\sum_i \mu_i x_i x_i^T = -X^T \operatorname{diag}(\mu_i) X.$$

Hence the observed information is

$$\boxed{\mathcal{I}_{\text{obs}}(\beta) = -\nabla^2 \ell(\beta) = X^T \operatorname{diag}(\mu_i) X.}$$

For the expected information, take expectation of the observed information. Since μ_i is a function of the parameter and design, not of the random response,

$$\mathcal{I}_{\text{exp}}(\beta) = \mathbb{E}\{X^T \operatorname{diag}(\mu_i) X\} = X^T \operatorname{diag}(\mu_i) X.$$

Thus the observed and expected information coincide exactly.

(b) Derive the weighted least-squares working response.

Newton's method for maximizing the log-likelihood is

$$\beta^{(k+1)} = \beta^{(k)} + \{X^T W^{(k)} X\}^{-1} X^T (y - \mu^{(k)}),$$

where $W^{(k)} = \operatorname{diag}(\mu_i^{(k)})$. Define the working response

$$z_i^{(k)} = \eta_i^{(k)} + \frac{y_i - \mu_i^{(k)}}{\mu_i^{(k)}}, \qquad \eta_i^{(k)} = x_i^T \beta^{(k)}.$$

Then

$$X^T W^{(k)} (z^{(k)} - X\beta^{(k)}) = X^T W^{(k)} \left(\frac{y - \mu^{(k)}}{\mu^{(k)}} \right) = X^T (y - \mu^{(k)}),$$

where division by $\mu^{(k)}$ is componentwise. Therefore the Newton update is the weighted least-squares solution

$$\boxed{\beta^{(k+1)} = (X^T W^{(k)} X)^{-1} X^T W^{(k)} z^{(k)},}$$

with

$$\boxed{w_i^{(k)} = \mu_i^{(k)}, \qquad z_i^{(k)} = \eta_i^{(k)} + \frac{y_i - \mu_i^{(k)}}{\mu_i^{(k)}}.}$$

Because observed and expected information are the same for this model, Newton and Fisher scoring have the same update.

(c) One iteration by hand.

The sample mean is

$$\overline{y} = \frac{2+3+6}{3} = \frac{11}{3}.$$

At $\beta^{(0)} = (\log(11/3), 0)$, all three linear predictors are $\eta_i^{(0)} = \log(11/3)$ and all fitted means are

$$\mu_i^{(0)} = \frac{11}{3}.$$

The design matrix has rows $(1,0), (1,1), (1,2)$. The score is

$$X^T(y - \mu^{(0)}) = \begin{pmatrix} 2+3+6-11 \\ 0(2-11/3) + 1(3-11/3) + 2(6-11/3) \end{pmatrix} = \begin{pmatrix} 0 \\ 4 \end{pmatrix}.$$

The information matrix is

$$X^T W X = \frac{11}{3}\begin{pmatrix} 3 & 3 \\ 3 & 5 \end{pmatrix}.$$

Let the update be $\Delta = (\Delta_0, \Delta_1)^T$. Solving

$$\frac{11}{3}\begin{pmatrix} 3 & 3 \\ 3 & 5 \end{pmatrix}\begin{pmatrix} \Delta_0 \\ \Delta_1 \end{pmatrix} = \begin{pmatrix} 0 \\ 4 \end{pmatrix}$$

first gives $\Delta_0 = -\Delta_1$. The second equation then becomes

$$\frac{11}{3}(-3\Delta_1 + 5\Delta_1) = 4 \quad \Longrightarrow \quad \Delta_1 = \frac{6}{11}, \quad \Delta_0 = -\frac{6}{11}.$$

Therefore

$$\boxed{\beta^{(1)} = \begin{pmatrix} \log(11/3) - 6/11 \\ 6/11 \end{pmatrix}.}$$

(d) Why there are no spurious local maxima.

The Hessian of the log-likelihood is

$$\nabla^2 \ell(\beta) = -X^T \operatorname{diag}(\mu_i) X.$$

For every vector v,

$$v^T \nabla^2 \ell(\beta) v = -\sum_i \mu_i (x_i^T v)^2 \le 0,$$

because every $\mu_i > 0$. Thus the log-likelihood is concave in β. A local maximum of a concave function is global; if X has full column rank, the Hessian is negative definite and the log-likelihood is strictly concave, so there is at most one finite maximizer. Nonlinear least-squares objectives do not generally have this concavity property because the nonlinear mean function can create multiple local minima.

(e) Equality of total observed and fitted counts.

At an interior MLE, the score is zero. The intercept column of X consists of ones, so the first component of the score equation is

$$0 = \sum_i (y_i - \widehat{\mu}_i).$$

Rearranging gives

$$\boxed{\sum_i y_i = \sum_i \widehat{\mu}_i.}$$

The fitted total count therefore equals the observed total count at the MLE of a model containing an intercept.

Key Takeaway

For Poisson log-linear regression, the observed and expected information coincide, the likelihood is concave, and Newton/Fisher scoring becomes weighted least squares of a working response.

10 Problem 10: One Gauss–Newton step and one Newton step by hand

Problem Statement

Data: $x = (1, 2)$, $y = (0.62, 0.18)$. Model: $f(x, \theta) = e^{-\theta x}$. Start: $\theta^{(0)} = 1$.

(a) Compute f_i, r_i, J_i and f_i'' at $\theta^{(0)}$.

(b) Compute one Gauss–Newton step $\theta_{GN}^{(1)}$.

(c) Compute one Newton step $\theta_N^{(1)}$.

(d) The minimizer is $\widehat{\theta} = 0.6489$. Evaluate S at $\theta^{(0)}$, $\theta_{GN}^{(1)}$, $\theta_N^{(1)}$ and $\widehat{\theta}$. Which step was better? Explain why, using the two quadratic models of S that the methods minimize.

(e) "Newton converges quadratically, so it is always better." Comment.

Required Theory

For $f_i(\theta) = f(x_i, \theta)$, define

$$r_i(\theta) = y_i - f_i(\theta), \qquad J_i(\theta) = f_i'(\theta), \qquad f_i''(\theta) = \frac{d^2 f_i}{d\theta^2}.$$

The Gauss–Newton step minimizes the least-squares objective after linearizing the mean function:

$$\theta_{GN}^{(k+1)} = \theta^{(k)} + \frac{\sum_i J_i r_i}{\sum_i J_i^2}.$$

The Newton step minimizes the second-order Taylor approximation to the actual sum of squares:

$$\theta_N^{(k+1)} = \theta^{(k)} + \frac{\sum_i J_i r_i}{\sum_i J_i^2 - \sum_i r_i f_i''},$$

provided the denominator (half the Hessian) is positive. These are different local quadratic approximations; neither formula guarantees that the actual objective decreases after a large step.

Solution

(a) Evaluate the fitted values and derivatives at $\theta^{(0)} = 1$.

For $f_i(\theta) = e^{-\theta x_i}$,

$$J_i(\theta) = -x_i e^{-\theta x_i}, \qquad f_i''(\theta) = x_i^2 e^{-\theta x_i}.$$

At $\theta = 1$:

	$i = 1$ $(x_1 = 1)$	$i = 2$ $(x_2 = 2)$
$f_i = e^{-x_i}$	0.367879	0.135335
$r_i = y_i - f_i$	0.252121	0.044665
$J_i = -x_i e^{-x_i}$	−0.367879	−0.270671
$f_i'' = x_i^2 e^{-x_i}$	0.367879	0.541341

The two sums used by the iterations are

$$\sum_i J_i r_i \approx -0.1048394, \qquad \sum_i J_i^2 \approx 0.2085978, \qquad \sum_i r_i f_i'' \approx 0.1169288.$$

(b) Gauss–Newton step.

Substitute the first two sums:

$$\theta_{GN}^{(1)} = 1 + \frac{-0.1048394}{0.2085978} \approx \boxed{0.4974090}.$$

(c) Newton step.

The denominator for Newton is

$$\sum_i J_i^2 - \sum_i r_i f_i'' \approx 0.2085978 - 0.1169288 = 0.0916690.$$

Therefore

$$\theta_{N}^{(1)} = 1 + \frac{-0.1048394}{0.0916690} \approx \boxed{-0.1436731}.$$

The denominator is positive, so the local second-order quadratic has a finite minimizer; nevertheless, that minimizer is far from the actual solution.

(d) Compare the actual objective values.

The objective is

$$S(\theta) = (0.62 - e^{-\theta})^2 + (0.18 - e^{-2\theta})^2.$$

Evaluate it at the four requested values:

parameter value	$S(\theta)$
$\theta^{(0)} = 1$	0.0655597
$\theta_{GN}^{(1)} \approx 0.4974090$	0.0361620
$\theta_{N}^{(1)} \approx -0.1436731$	1.6148427
$\widehat{\theta} = 0.6489$	0.0181564

The Gauss–Newton step is better for this particular starting value: it reduces the actual sum of squares, while the Newton step increases it dramatically.

Why can Newton perform worse here? Gauss–Newton replaces the nonlinear mean by its first-order linearization, producing a quadratic approximation based on $J^T J$. Newton uses the exact second derivative of the sum of squares at the current point, including the residual-curvature correction $-\sum_i r_i f_i''$. That correction makes Newton's local quadratic more faithful to the curvature at the current point, but a quadratic approximation is only local. Its minimizer can lie

far away, where the approximation is no longer reliable. In this example it points to a negative value of θ, at which the actual objective is much larger.

(e) Why quadratic convergence does not make Newton always better.

Quadratic convergence is a local statement: under regularity conditions and a sufficiently good starting value, Newton's error can be approximately squared at each iteration. It is not a global guarantee. A Newton step can be too large, leave the useful region, encounter a non-positive Hessian, or move to a point with a larger objective. Damping, line search, or a better initial value may be needed. The present numerical example directly shows that one Newton step can be worse than one Gauss–Newton step even when the Hessian-based update is well-defined.

Key Takeaway

The most accurate local quadratic model does not necessarily give the best finite step. Always evaluate the actual objective and remember that Newton's fast convergence guarantee is local, not global.

11 Problem 11: Local convergence speed of Gauss–Newton

Problem Statement

For a one-parameter model, Gauss–Newton is the iteration $\theta^{(k+1)} = G(\theta^{(k)})$ with

$$G(\theta) = \theta + \frac{\sum_i J_i(\theta) r_i(\theta)}{\sum_i J_i(\theta)^2}.$$

(a) Show that

$$G'(\widehat{\theta}) = \frac{\sum_i f_i''(\widehat{\theta}) r_i(\widehat{\theta})}{\sum_i J_i(\widehat{\theta})^2}.$$

(b) Deduce the local rate of convergence. When is it faster than linear?

(c) What does the sign of $G'(\widehat{\theta})$ tell you about the iterates?

(d) For the data of Problem 10, compute $G'(\widehat{\theta})$ and compare it with the observed error ratios e_{k+1}/e_k, where $e_k = \theta^{(k)} - \widehat{\theta}$:

$$0.041, \quad -0.083, \quad -0.0895, \quad -0.0890, \quad -0.0890.$$

(e) Can Gauss–Newton fail to converge even when started very close to $\widehat{\theta}$?

Required Theory

A fixed point $\widehat{\theta}$ of an iteration satisfies $G(\widehat{\theta}) = \widehat{\theta}$. If G is differentiable near the fixed point, a first-order Taylor expansion gives

$$e_{k+1} = G(\theta^{(k)}) - G(\widehat{\theta}) \approx G'(\widehat{\theta}) e_k.$$

Thus the magnitude of $G'(\widehat{\theta})$ controls the local linear convergence factor. If the derivative is zero, the linear error term disappears and convergence is faster than linear, subject to the smoothness of the iteration map.

Solution

(a) Differentiate the Gauss–Newton map.

Write

$$A(\theta) = \sum_i J_i(\theta) r_i(\theta), \qquad B(\theta) = \sum_i J_i(\theta)^2,$$

so $G(\theta) = \theta + A(\theta)/B(\theta)$. Since $r_i' = -J_i$ and $J_i' = f_i''$,

$$A'(\theta) = \sum_i \{f_i'' r_i - J_i^2\}, \qquad B'(\theta) = 2\sum_i J_i f_i''.$$

At a least-squares solution, the score equation is $A(\widehat{\theta}) = \sum_i J_i r_i = 0$. Differentiate the quotient:

$$G'(\widehat{\theta}) = 1 + \frac{A'(\widehat{\theta})B(\widehat{\theta}) - A(\widehat{\theta})B'(\widehat{\theta})}{B(\widehat{\theta})^2}.$$

The term containing $A(\widehat{\theta})$ is zero. Therefore

$$G'(\widehat{\theta}) = 1 + \frac{\sum_i f_i''(\widehat{\theta}) r_i(\widehat{\theta}) - \sum_i J_i(\widehat{\theta})^2}{\sum_i J_i(\widehat{\theta})^2}$$
$$= \boxed{\frac{\sum_i f_i''(\widehat{\theta}) r_i(\widehat{\theta})}{\sum_i J_i(\widehat{\theta})^2}}.$$

(b) Local convergence rate.

Since $G(\widehat{\theta}) = \widehat{\theta}$, Taylor expansion around the fixed point gives

$$e_{k+1} = G(\widehat{\theta} + e_k) - G(\widehat{\theta}) = G'(\widehat{\theta}) e_k + o(e_k).$$

If $0 < |G'(\widehat{\theta})| < 1$, the method converges locally linearly with asymptotic error ratio $|G'(\widehat{\theta})|$. If $|G'(\widehat{\theta})| > 1$, the fixed point is locally repelling. If $G'(\widehat{\theta}) = 0$, the first-order term vanishes, giving superlinear convergence; with sufficient higher-order differentiability, the next nonzero Taylor term determines a quadratic or faster rate.

(c) Sign of the derivative.

For sufficiently small errors,

$$e_{k+1} \approx G'(\widehat{\theta}) e_k.$$

If $G'(\widehat{\theta}) > 0$, the next error has the same sign as the current error, so the iterates approach from the same side (locally, without alternating across the solution). If $G'(\widehat{\theta}) < 0$, the error changes sign at successive steps, so the iterates alternate around the solution. The magnitude determines whether the alternating or non-alternating sequence contracts.

(d) Numerical derivative for Problem 10.

Problem 10 gives $f_i = e^{-\widehat{\theta} x_i}$, $J_i = -x_i e^{-\widehat{\theta} x_i}$, and $f_i'' = x_i^2 e^{-\widehat{\theta} x_i}$. At $\widehat{\theta} = 0.6489$,

	$x_1 = 1$	$x_2 = 2$
f_i	0.522620	0.273132
$r_i = y_i - f_i$	0.097380	−0.093132
J_i	−0.522620	−0.546264
f_i''	0.522620	1.092528

Consequently,

$$G'(\widehat{\theta}) = \frac{\sum_i f_i'' r_i}{\sum_i J_i^2} \approx \boxed{-0.088983}.$$

The observed error ratios in the question settle at approximately -0.0890. This agrees extremely closely with the theoretical local derivative. The negative sign explains the alternating errors, and the absolute value about 0.089 explains the rapid contraction.

(e) Can Gauss–Newton fail even very near the solution?

Yes, in principle. A fixed point is locally attracting only when

$$|G'(\widehat{\theta})| < 1.$$

If $|G'(\widehat{\theta})| > 1$, sufficiently small nonzero errors are magnified rather than reduced, so starting close to the solution does not rescue the iteration. If the derivative has magnitude one, higher-order terms determine the behavior and convergence is not guaranteed. In the specific Problem 10 example, however, $|G'(\widehat{\theta})| \approx 0.089 < 1$, so the fixed point is strongly locally attracting.

Key Takeaway

Gauss–Newton's local rate is governed by the residual-curvature term $\sum_i f_i'' r_i$ relative to $\sum_i J_i^2$. Small residual-curvature at the solution means fast convergence; a derivative of magnitude above one means local instability.

12 Problem 12: Three Wald tests and an extra-sum-of-squares test

Problem Statement

Theophylline data, subject 1 ($n = 11$ observations, $p = 3$ parameters; use the t_8 distribution). The one-compartment fit, parametrized by $(\log k_e, \log k_a, \log CL)$, gives

$$\widehat{\log k_e} = -2.9196, \qquad \mathrm{se}(\widehat{\log k_e}) = 0.1709, \qquad S(\widehat{\theta}) = 4.2860.$$

We test $H_0 : k_e = 0.08$ per hour; equivalently, $\log k_e = \log 0.08$; equivalently, the half-life is $t_{1/2} = \log 2/0.08 = 8.66$ hours. Use $t_{8,0.975} = 2.306$ and $F_{1,8,0.95} = 5.318$.

(a) Compute the Wald t statistic on the $\log k_e$ scale, and state your decision at the 5% level.

(b) Use the delta method to obtain $\widehat{k}_e$, its standard error, and the Wald test on the k_e scale.

(c) Do the same on the half-life scale.

(d) Show the exact identity $t_{\text{half-life scale}} = -t_{k_e \text{ scale}}\, \widehat{k}_e/k_0$, where $k_0 = 0.08$.

(e) Refitting with k_e fixed at 0.08 gives $S_R = 7.5678$. Compute the extra-sum-of-squares F statistic. Why can its value not depend on the scale?

Required Theory

A Wald statistic compares an estimate to its null value using an estimated standard error on the chosen parameter scale. The delta method approximates the standard error of a smooth transformed estimator by multiplying the original standard error by the absolute derivative of the transformation at the estimate. Wald statistics are not generally invariant under nonlinear reparameterization because the approximation is local and symmetric on the scale on which it is constructed.

For a restriction of one parameter in nonlinear least squares, the extra-sum-of-squares statistic is

$$F = \frac{(S_R - S_F)/q}{S_F/(n-p)},$$

where S_R and S_F are the restricted and full residual sums of squares and q is the number of restrictions. A one-to-one change of parameter scale does not change the restricted or full model sets, so this objective-based test is scale-invariant.

Solution

(a) Wald test on the log scale.

The null value is

$$\log k_0 = \log(0.08) \approx -2.525729.$$

The Wald statistic is

$$t_{\log k_e} = \frac{-2.9196 - (-2.525729)}{0.1709} \approx \boxed{-2.30469}.$$

Since $|t| = 2.30469 < 2.306$, it falls just below the supplied 5% two-sided critical value. We therefore do not reject H_0 at the 5% level on the log scale. The decision is extremely close to the cutoff.

(b) Wald test on the k_e scale.

Transform the estimate:

$$\widehat{k}_e = e^{-2.9196} \approx \boxed{0.0539553}.$$

For $g(\eta) = e^{\eta}$, $g'(\eta) = e^{\eta}$. The delta method gives

$$\operatorname{se}(\widehat{k}_e) \approx e^{\widehat{\log k_e}} \operatorname{se}(\widehat{\log k_e}) = 0.0539553(0.1709) \approx \boxed{0.0092210}.$$

The Wald statistic on the original rate scale is

$$t_{k_e} = \frac{0.0539553 - 0.08}{0.0092210} \approx \boxed{-2.824}.$$

Its absolute value exceeds 2.306, so this scale's Wald test rejects at 5%. This differs from part (a), even though the null hypothesis is the same.

(c) Wald test on the half-life scale.

The half-life is $h(k_e) = \log 2/k_e$. Therefore

$$\widehat{t}_{1/2} = \frac{\log 2}{0.0539553} \approx \boxed{12.8467 \text{ hours}}.$$

The derivative is $h'(k) = -\log 2/k^2$. The delta-method standard error is

$$\begin{aligned}
\operatorname{se}(\widehat{t}_{1/2}) &\approx \frac{\log 2}{\widehat{k}_e^2} \operatorname{se}(\widehat{k}_e) \\
&= \widehat{t}_{1/2} \operatorname{se}(\widehat{\log k_e}) \\
&\approx 12.8467(0.1709) = \boxed{2.1955 \text{ hours}}.
\end{aligned}$$

The null half-life is $t_0 = \log 2/0.08 \approx 8.6643$ hours. Hence

$$t_{1/2 \text{ scale}} = \frac{12.8467 - 8.6643}{2.1955} \approx \boxed{1.9050}.$$

Since $1.9050 < 2.306$, we do not reject at the 5% level on the half-life scale.

(d) Prove the exact identity.

Let $k_0 = 0.08$. The delta-method standard error on the k scale is $\widehat{k}_e s_\eta$, where $s_\eta = \widehat{\text{se}(\log k_e)}$. Thus

$$t_{k_e \text{ scale}} = \frac{\widehat{k}_e - k_0}{\widehat{k}_e s_\eta}.$$

On the half-life scale, the estimate is $\log 2/\widehat{k}_e$, the null value is $\log 2/k_0$, and the delta-method standard error is $(\log 2/\widehat{k}_e)s_\eta$. Therefore

$$\begin{aligned} t_{\text{half-life scale}} &= \frac{\log 2/\widehat{k}_e - \log 2/k_0}{(\log 2/\widehat{k}_e)s_\eta} \\ &= \frac{1 - \widehat{k}_e/k_0}{s_\eta} \\ &= -\frac{\widehat{k}_e}{k_0}\frac{\widehat{k}_e - k_0}{\widehat{k}_e s_\eta} \\ &= \boxed{-t_{k_e \text{ scale}}\frac{\widehat{k}_e}{k_0}}. \end{aligned}$$

Numerically, $-(-2.824)(0.0539553/0.08) \approx 1.905$, as obtained in part (c).

(e) Extra-sum-of-squares test.

There is one restriction, so $q = 1$. The full model has $n - p = 11 - 3 = 8$ residual degrees of freedom. Therefore

$$\begin{aligned} F &= \frac{(7.5678 - 4.2860)/1}{4.2860/8} \\ &= \frac{3.2818}{0.53575} \approx \boxed{6.1256}. \end{aligned}$$

Since $6.1256 > 5.318$, the extra-sum-of-squares test rejects the null at 5%.

Why is this test independent of the scale? The full model and the model with $k_e = 0.08$ describe the same two sets of mean functions whether the rate is represented by k_e, $\log k_e$, or the half-life. A one-to-one reparameterization changes the coordinates used to describe those sets, but it does not change the fitted values, S_F, or S_R. Therefore the extra-sum-of-squares F statistic is unchanged.

Interpretation of the disagreement. The three Wald statistics are local symmetric approximations on three different scales; nonlinear transformations distort symmetry and standard errors. Their decisions need not agree in a nonlinear model. The objective-based extra-sum-of-squares test compares the same restricted and unrestricted fits and is invariant to the parameter scale. Here it rejects, while the Wald decisions differ by scale.

Key Takeaway

Wald tests are parameterization-dependent in nonlinear models. The extra-sum-of-squares test is based on fitted-model comparisons and is invariant under a one-to-one change of parameter scale.

13 Problem 13: Why profile and Wald intervals agree in linear models

Problem Statement

(a) In the linear model $y = X\beta + \varepsilon$ (X of full column rank), show that for every b,

$$\widetilde{S}(b) - S(\widehat{\beta}) = \frac{(b - \widehat{\beta}_j)^2}{[(X^T X)^{-1}]_{jj}}, \qquad \widetilde{S}(b) = \min_{\beta:\beta_j = b} S(\beta).$$

Deduce that the profile t function

$$\tau(b) = \operatorname{sign}(b - \widehat{\beta}_j)\frac{\sqrt{\widetilde{S}(b) - S(\widehat{\beta})}}{s}$$

equals $(b - \widehat{\beta}_j)/\operatorname{se}(\widehat{\beta}_j)$, a straight line.

(b) In a nonlinear model, let $\phi = g(\theta_j)$ with g strictly increasing. Show that the profile interval for ϕ is exactly g(profile interval for θ_j). Is the same true for Wald intervals?

(c) Explain why curvature in the plot of $\tau(\theta_j)$ against θ_j is a diagnostic of non-linearity.

Required Theory

In a full-rank linear model, the least-squares objective is a quadratic function of the coefficient vector. Its constrained minimum as one coefficient is fixed is also quadratic in the fixed value. The diagonal element $[(X^T X)^{-1}]_{jj}$ is the variance factor for that coefficient. A profile interval is defined by the set of parameter values whose constrained objective increase does not exceed a specified cutoff. A monotone transformation maps this set one-to-one, but a Wald interval is built from a local normal approximation and is not generally invariant under nonlinear transformations.

Solution

(a) Derive the profile sum of squares identity.

Let $C = (X^T X)^{-1}$, and let e_j be the jth coordinate vector. The unconstrained OLS estimator satisfies

$$X^T(y - X\widehat{\beta}) = 0.$$

We seek the coefficient vector minimizing $S(\beta) = \|y - X\beta\|^2$ subject to $e_j^T\beta = b$. Write $\Delta = b - \widehat{\beta}_j$. A Lagrange-multiplier calculation gives the constrained minimizer

$$\widetilde{\beta}(b) = \widehat{\beta} + \frac{\Delta}{C_{jj}} C e_j.$$

Indeed, its jth coordinate is

$$\widehat{\beta}_j + \frac{\Delta}{C_{jj}} e_j^T C e_j = \widehat{\beta}_j + \Delta = b.$$

The constrained displacement is $d = \widetilde{\beta} - \widehat{\beta} = (\Delta/C_{jj})Ce_j$. Since the OLS residual is orthogonal to the columns of X,

$$S(\widehat{\beta} + d) - S(\widehat{\beta}) = d^T X^T X d.$$

Substitute d:

$$\begin{aligned} d^T X^T X d &= \frac{\Delta^2}{C_{jj}^2} e_j^T C^T X^T X C e_j \\ &= \frac{\Delta^2}{C_{jj}^2} e_j^T C e_j = \frac{\Delta^2}{C_{jj}}. \end{aligned}$$

Therefore

$$\boxed{\widetilde{S}(b) - S(\widehat{\beta}) = \frac{(b - \widehat{\beta}_j)^2}{[(X^T X)^{-1}]_{jj}}.}$$

The usual estimated standard error is

$$\mathrm{se}(\widehat{\beta}_j) = s\sqrt{[(X^T X)^{-1}]_{jj}}.$$

Using the profile identity,

$$\begin{aligned} \tau(b) &= \mathrm{sign}(b - \widehat{\beta}_j) \frac{|b - \widehat{\beta}_j|}{s\sqrt{[(X^T X)^{-1}]_{jj}}} \\ &= \boxed{\frac{b - \widehat{\beta}_j}{\mathrm{se}(\widehat{\beta}_j)}}. \end{aligned}$$

This is a straight line in b, passing through zero at $b = \widehat{\beta}_j$.

(b) Profile intervals under a monotone transformation.

Let $\phi = g(\theta_j)$, where g is strictly increasing. Because g is one-to-one, fixing ϕ at a value u is equivalent to fixing θ_j at $g^{-1}(u)$. The remaining parameters are profiled out in exactly the same way, so the profile objective at u is the original profile objective evaluated at $g^{-1}(u)$. Consequently the set of accepted profile values transforms exactly:

$$\boxed{\mathcal{I}_\phi = g(\mathcal{I}_{\theta_j}).}$$

In words, transform both endpoints of the profile interval by g; because g is increasing, their order is preserved.

The same is not generally true for Wald intervals. A Wald interval for θ_j is approximately symmetric around $\widehat{\theta}_j$. Applying a nonlinear g produces an interval that is generally asymmetric around $g(\widehat{\theta}_j)$. A Wald interval constructed directly on the ϕ scale instead uses a delta-method standard error $|g'(\widehat{\theta}_j)|\,\mathrm{se}(\widehat{\theta}_j)$ and is symmetric on the ϕ scale. These two constructions agree only approximately for transformations that are nearly linear over the interval (or exactly for affine transformations).

(c) Curvature as a diagnostic.

In a linear model, part (a) shows that the signed profile statistic $\tau(\theta_j)$ is exactly linear in θ_j. In a nonlinear model, the profile sum of squares need not be quadratic in the parameter, so the graph of the signed square root of the profile increase can bend. Such curvature indicates that the objective changes at different rates on opposite sides of the estimate, or that the local quadratic approximation is inadequate over the interval. It is therefore a diagnostic of nonlinear parameter geometry and a warning that a symmetric Wald interval may not describe the profile-based uncertainty well.

Key Takeaway

Linear least squares produces an exactly quadratic profile and a straight profile-t curve. Curvature in a nonlinear profile reveals departures from that simple quadratic geometry, while profile intervals retain exact transformation behavior that Wald intervals generally lack.

14 Problem 14: Optimal design points for two nonlinear models

Problem Statement

(a) Model $\theta_1 e^{-\theta_2 x}$ ($\theta_1, \theta_2 > 0$), with $n/2$ observations at $x = 0$ and $n/2$ at $x = c > 0$. Show that

$$\mathrm{Var}(\widehat{\theta}_2) \approx \frac{2\sigma^2\theta_2^2}{n\theta_1^2} \cdot \frac{e^{2u}+1}{u^2}, \qquad u = \theta_2 c,$$

and that the best c solves $u = 1 + e^{-2u}$. Solve this equation by the fixed-point iteration $u^{(k+1)} = 1 + e^{-2u^{(k)}}$ from $u^{(0)} = 1$ (three or four steps suffice), and express the optimal c in half-lives. Why can we not use this design directly?

(b) Michaelis–Menten $V_m x/(K + x)$, with $n/2$ observations at a and $n/2$ at b, where $0 < a < b$ and b is the largest feasible concentration. Show that $\det(J^T J) \propto \{a(b-a)/(K+a)^2\}^2$, and that the D-optimal choice (maximizing $\det(J^T J)$) is

$$a^* = \frac{Kb}{2K+b}.$$

Evaluate it for $K = 0.0641$, $b = 1.10$ (Puromycin). What happens as $a \to b$? As $a \to 0$?

Required Theory

For a nonlinear mean vector $f(\theta)$ with Jacobian matrix J, the local covariance approximation supplied in the problem sheet is

$$\mathrm{Cov}(\widehat{\theta}) \approx \sigma^2 (J^T J)^{-1}.$$

For a two-parameter matrix

$$J^T J = \begin{pmatrix} A & B \\ B & C \end{pmatrix}, \qquad (J^T J)^{-1}_{22} = \frac{A}{AC - B^2}.$$

The determinant $\det(J^T J)$ measures the area information in the two-parameter Jacobian. A D-optimal design maximizes this determinant. In a local design calculation, the optimal point generally depends on unknown parameter values and is therefore only locally optimal at the parameter values used to compute it.

Solution

(a) Exponential-decay design.

The mean is $f(x) = \theta_1 e^{-\theta_2 x}$. Its two derivatives are

$$\frac{\partial f}{\partial \theta_1} = e^{-\theta_2 x}, \qquad \frac{\partial f}{\partial \theta_2} = -\theta_1 x e^{-\theta_2 x}.$$

At $x = 0$, a Jacobian row is $(1, 0)$. At $x = c$, it is

$$(e^{-\theta_2 c}, -\theta_1 c e^{-\theta_2 c}).$$

There are $n/2$ observations at each point. Therefore

$$J^T J = \frac{n}{2} \begin{pmatrix} 1 + e^{-2\theta_2 c} & -\theta_1 c e^{-2\theta_2 c} \\ -\theta_1 c e^{-2\theta_2 c} & \theta_1^2 c^2 e^{-2\theta_2 c} \end{pmatrix}.$$

Write $u = \theta_2 c$. The determinant of the matrix inside the factor $n/2$ is

$$\begin{aligned} &(1 + e^{-2u})\theta_1^2 c^2 e^{-2u} - \theta_1^2 c^2 e^{-4u} \\ &= \theta_1^2 c^2 e^{-2u}. \end{aligned}$$

The upper-left entry of $J^T J$ is $(n/2)(1 + e^{-2u})$. Hence the $(2, 2)$ element of its inverse is

$$\begin{aligned} \{(J^T J)^{-1}\}_{22} &= \frac{(n/2)(1 + e^{-2u})}{(n/2)^2 \theta_1^2 c^2 e^{-2u}} \\ &= \frac{2}{n\theta_1^2 c^2}(e^{2u} + 1). \end{aligned}$$

Multiplying by σ^2 and using $c = u/\theta_2$ gives

$$\boxed{\mathrm{Var}(\widehat{\theta}_2) \approx \frac{2\sigma^2 \theta_2^2}{n\theta_1^2} \frac{e^{2u} + 1}{u^2}.}$$

Only the final factor depends on the design variable u. Minimize

$$q(u) = \frac{e^{2u} + 1}{u^2}, \qquad u > 0.$$

Differentiate its logarithm:

$$\frac{d}{du} \log q(u) = \frac{2e^{2u}}{e^{2u} + 1} - \frac{2}{u}.$$

At an interior optimum this equals zero, so

$$\frac{ue^{2u}}{e^{2u} + 1} = 1 \quad \Longleftrightarrow \quad u = 1 + e^{-2u}.$$

The fixed-point iterations from $u^{(0)} = 1$ are

$$\begin{aligned} u^{(1)} &= 1 + e^{-2} = 1.135335, \\ u^{(2)} &= 1 + e^{-2(1.135335)} = 1.103243, \\ u^{(3)} &= 1 + e^{-2(1.103243)} = 1.110087, \\ u^{(4)} &= 1 + e^{-2(1.110087)} = 1.108590. \end{aligned}$$

The fixed point is approximately $u^* = 1.10886$. Thus

$$c^* = \frac{u^*}{\theta_2}.$$

The half-life is $t_{1/2} = \log 2/\theta_2$, so

$$\frac{c^*}{t_{1/2}} = \frac{u^*}{\log 2} \approx \boxed{1.599 \text{ half-lives.}}$$

We cannot apply this design directly without qualification because the optimal c^* depends on the unknown rate θ_2. A pilot estimate or prior value is needed to choose a locally optimal design; the design can then be updated as information about θ_2 improves.

(b) Michaelis–Menten D-optimal design.

For

$$f(x; V_m, K) = \frac{V_m x}{K + x},$$

the Jacobian row is

$$J(x) = \left(\frac{x}{K + x}, -\frac{V_m x}{(K + x)^2} \right).$$

With $n/2$ replicates at a and $n/2$ at b, the determinant of $J^T J$ is proportional to the square of the determinant of the two distinct Jacobian rows. That two-row determinant is

$$\det \begin{pmatrix} \frac{a}{K+a} & -\frac{V_m a}{(K+a)^2} \\ \frac{b}{K+b} & -\frac{V_m b}{(K+b)^2} \end{pmatrix} = \frac{V_m ab(b-a)}{(K+a)^2(K+b)^2}.$$

Consequently

$$\det(J^T J) \propto \left\{ \frac{ab(b-a)}{(K+a)^2(K+b)^2} \right\}^2.$$

For fixed b and K, the factor $b^2/(K+b)^4$ is constant with respect to a. Thus maximizing the determinant is equivalent to maximizing

$$q(a) = \frac{a(b-a)}{(K+a)^2}, \qquad 0 < a < b.$$

Differentiate its logarithm:

$$\frac{d}{da} \log q(a) = \frac{1}{a} - \frac{1}{b-a} - \frac{2}{K+a}.$$

Set this equal to zero and clear the positive denominators:

$$(b-a)(K+a) - a(K+a) - 2a(b-a) = 0.$$

Simplifying yields

$$Kb - (2K + b)a = 0,$$

so

$$\boxed{a^* = \frac{Kb}{2K+b}.}$$

For $K = 0.0641$ and $b = 1.10$,

$$a^* = \frac{(0.0641)(1.10)}{2(0.0641) + 1.10} \approx \boxed{0.05741}.$$

As $a \to b$, the two design points merge. The two Jacobian rows become nearly identical, $b - a \to 0$, and the determinant tends to zero: the design provides little ability to distinguish the two parameters. As $a \to 0$, the response and both derivative components at the lower concentration approach zero, so that point contributes almost no information about the parameters; the determinant again tends to zero. The optimum lies strictly between the endpoints and balances sensitivity at a low concentration with separation from the high concentration.

Key Takeaway

The local optimal design balances parameter sensitivity across design points. Because the optimum depends on unknown parameters, it is a local design and generally needs a pilot estimate; concentrating at nearly identical points destroys the information determinant.

A Notation and algorithm reference

For nonlinear least squares,

$$S(\theta) = \sum_i r_i(\theta)^2, \qquad r_i(\theta) = y_i - f(x_i, \theta), \qquad J_{ij}(\theta) = \frac{\partial f(x_i, \theta)}{\partial \theta_j}.$$

The gradient and Hessian are

$$\nabla S(\theta) = -2J^T r, \qquad \nabla^2 S(\theta) = 2\left(J^T J - \sum_i r_i \nabla^2 f(x_i, \theta)\right).$$

The three update ideas used in this problem set are:

$$\begin{aligned} \text{Gauss–Newton: } & \theta^{(k+1)} = \theta^{(k)} + (J^T J)^{-1} J^T r, \\ \text{Newton: } & \theta^{(k+1)} = \theta^{(k)} - \{\nabla^2 S(\theta^{(k)})\}^{-1} \nabla S(\theta^{(k)}), \\ \text{Fisher scoring: } & \theta^{(k+1)} = \theta^{(k)} + I(\theta^{(k)})^{-1} \nabla \ell(\theta^{(k)}). \end{aligned}$$

For normal-error nonlinear regression, Fisher scoring has the Gauss–Newton form. In the one-parameter case, the Newton and Gauss–Newton formulas are shown separately in the theory sections of Problems 3 and 10.

B Scope of the supplied materials

The supplied PS4 sheet contains the nonlinear-regression setup, the residual/Jacobian definitions, and the Gauss–Newton, Newton, and Fisher-scoring formulas used in these solutions. The attached file is titled *ASM II pre-midsem notes*; its searchable content includes a section on LAD/MM robust regression but does not include a nonlinear-regression lecture. Accordingly, the nonlinear-regression theory in this document is taken from the PS4 sheet itself and is developed directly in the individual solutions.